

The C -threshold

Frame anatomy of the μ -dual CD Gram on Y , and why Nyquist-at-density- $\frac{1}{2}$ is not the theorem

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Abstract

Round 407 reduced P1 to $\lambda_2(C) \geq 1$, with C the μ -only dual Christoffel–Darboux Gram on the hole nodes Y . The candidate lemma of this round is that C has at most one eigenvalue below 1 because half-filling is Nyquist-critical. That lemma is refuted as a theorem: source hole density is 0.28 not $\frac{1}{2}$; the periodic 1010 tile at density $\frac{1}{2}$ has a kernel and three or four eigenvalues at 0; $\text{diag}(C)$ is far above 1. Finite linear-algebra identities remain: 2×2 coherence, Rayleigh, rank deficiency, Cauchy interlacing. On the source family, $\#\{\lambda(C) < 1\} \in \{0, 1\}$ is a census (42/42), not a proof. No RH claim.

Status. Lemmas 1–4 are proved (finite identities over $\mathbb{Q}$). The Nyquist-at-density- $\frac{1}{2}$ candidate is **refuted**. Propositions 2–4 are a finite machine census on FRAME-A, core-42, and the χ_3 twins. No RH claim. No L^* claim. No $\lambda_2(C) \geq 1$ claim.

1 The object

Write $u^\vee$ for the Borodin dual of $|u|$ (r356) and $n = N_w - 3$. Let $P_0, \dots, P_{n-1}$ be the $u^\vee|_\mu$ -orthonormal orthogonal polynomials and $B_{ki} = \sqrt{u^\vee(y_k)} P_i(y_k)$. Then

$$C = BB^T, \quad C_{ij} = \sqrt{u^\vee(y_i) u^\vee(y_j)} K_n^\mu(y_i, y_j), \quad (1)$$

the dual CD kernel of the positive μ -chain, evaluated on Y and dressed by $\sqrt{u^\vee|_Y}$. Round 407: $R_{N-3} = C(I + C)^{-1}$ and $\text{ind}_- A_0 = \#\{\lambda(C) < 1\}$, so $\text{P1} \iff \lambda_2(C) \geq 1$.

2 Finite identities

Lemma 1 (Two-by-two coherence). *For $C = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ over $\mathbb{Q}$ with $a = 2$, $b = 3$, $c = 2$, $\det(C - I) = -2 < 0$. Hence $C - I$ has eigenvalues of opposite sign, so exactly one eigenvalue of C lies below 1. In general $\det(C - I) = (a - 1)(b - 1) - c^2$, so the 1-threshold on a 2×2 Gram is a coherence bound, not a diagonal bound.*

Lemma 2 (Rayleigh). $\lambda_{\min}(C) \leq \min_i C_{ii}$. In particular $C_{ii} < 1$ for some i forces $\#\{\lambda(C) < 1\} \geq 1$.

Lemma 3 (Rank deficiency). *If B is $|Y| \times n$ then $\text{rank } C \leq \min(|Y|, n)$. Over $\mathbb{Q}$, a 3×2 example has $\det C = 0$. Thus $|Y| > n$ produces at least $|Y| - n$ zeros, all strictly below 1.*

Lemma 4 (Cauchy interlacing). *A principal submatrix C_I of a Hermitian C satisfies $\lambda_{\min}(C_I) \geq \lambda_{\min}(C)$. Deleting hole coordinates of a frozen C cannot create new eigenvalues below 1; it can only raise $\lambda_{\min}$.*

3 Why Nyquist-at- $\frac{1}{2}$ fails

The half-filling slogan $S = 2N_w - 1$ is a statement about the cosine grid versus CD depth, not about $|Y|/S$. On FRAME-A, $|Y|/S = 104/367 = 0.283$ and $n = 181 > |Y|$, so C is typically full rank. The periodic 1010 tile (even/odd split, density $\frac{1}{2}$) has $n < |Y|$ and therefore a kernel.

Proposition 1 (Periodic 1010 is not $\lambda_2 = 1$). *On the two-period mesh of r367, $S = 20$ and $c = 1$ give density $\frac{1}{2}$, $\#\{\lambda(C) < 1\} = 3$ and $\lambda_{\min} = 0$ (at least two exact zeros). $S = 21$, $c = \frac{2}{3}$ gives four eigenvalues at 0. This is Lemma 3, not a critical frame with exactly one soft mode.*

Proposition 2 (FRAME-A anatomy). *On window $kz = 9$: $C_{\min} = 0.85712$, $\lambda_2(C) = 1.00018$, $\#\{\lambda < 1\} = 1$. Every diagonal entry exceeds 1.65 (mean 33.1). The exceptional eigenvector is not the ones-mode ($|\langle v, 1 \rangle| = 0.007$); it is hole-Nyquist ($k_{\text{peak}} = |Y|/2 = 52$) with endpoint mass 0.53.*

Thinning Y to 30% raises $C_{\min}$ above 1 (rebuilt, and also frozen-Cauchy). Flipping ten μ -atoms into holes produces eight eigenvalues below 1. The defect is not caused by too few holes.

4 Worlds and census

Proposition 3 (Named exclusion). *Scramble: $\#\{\lambda < 1\} = 21$, $\min \text{diag}(C) = 0.37 < 1$, fifteen adjacent-hole gaps — $d_{\min} < 1$ plus F1-clusters, at lower density than the source. Permute: the same Y as the source, but $\#\{\lambda < 1\} = 20$ and $d_{\min} = 0.15$; the dual weights, not the hole mask, break $d_{\min} > 1$. Live χ_3 ($kz = 9$): $\#\{\lambda < 1\} = 0$. Dead χ_3 ($kz = 15$): $\#\{\lambda < 1\} = 1$ — the threshold dictionary holds; death remains the edge signature.*

Proposition 4 (Source census). *On the admissible core of 42 windows, $\#\{\lambda(C) < 1\} \in \{0, 1\}$ (fourteen vacuous, twenty-eight with one), never greater; $\min \text{diag}(C) \in [1.097, 1.656]$. Every one-defect window has hole-Nyquist peak $k_{\text{peak}}/(|Y|/2) \geq 0.85$. Every vacuous window has endpoint mass 1 on the lowest mode (DC sitting at or above 1). This is a census of the source family, not a sampling theorem, and not an RH claim.*

Remark 1. P1 remains equivalent to $\lambda_2(C) \geq 1$ by r407. The present round does not prove the right-hand side. F1-run and gap-histogram statistics do not control $C_{\min}$ on the one-defect subsample (inherited from r393/r395: those coordinates do not carry). Unconditional SATZ is Lemmas 1–4. The census is source-weight-adjacent and is not gated as PNT or RH-near.

Kill table (machine)

mutant	$\#\{\lambda(C) < 1\}$	what breaks
1010 $S = 20$	3 (zeros)	$n < Y $ kernel
two-period $S = 21$	4	same
scramble	21	$d_{\min} < 1$, F1-clusters
Λ -permute	20	dual weights, same Y
thin Y to 30%	0	$C_{\min}$ rises
flip +10 $\mu \rightarrow Y$	8	extra holes
depth $n + 1$	1	razor moves to 0.99996